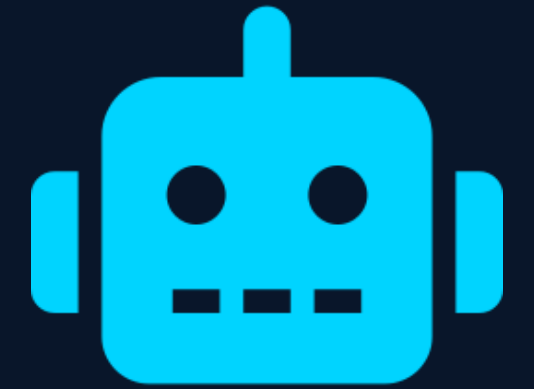


TEAM MSC-MICROSOFT STUDENT CLUB



CYBER CYPHER 5.0



TEAM - B103
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PROBLEM STATEMENT 1

Autonomous Network Operations Center Agent

The Challenge:

Regional ISPs manage massive, complex networks generating continuous telemetry — latency spikes, packet loss, routing flaps, congestion signals, and hardware degradation. Small anomalies compound silently across layers until they cascade into major outages. By then, SLAs are already breached and operations teams are stuck reacting instead of preventing.

The Agent Must:

- Observe: Ingest live network telemetry: latency, packet loss, utilization, routing changes, device health
- Reason: Detect early degradation signals and form root-cause hypotheses across network layers
- Decide: Balance service reliability, blast radius, and customer impact before acting
- Act: Execute traffic rerouting, rate limiting, config rollbacks, or escalate to engineers with guardrails
- Learn: Improve detection accuracy and decision confidence from past intervention outcomes

What It Is, Who It's For & Why It Matters



What It Is

An agentic AI system that autonomously monitors, diagnoses, and remediates network incidents across a simulated ISP backbone — powered by Claude, chaos engineering, and a 4-tier human-in-the-loop autonomy framework.



Who It's For

NOC Engineers

— fatigued by alert storms

ISP Operators

— managing high-availability backbone

AI/Platform Teams

— exploring autonomous agentic systems



Why It Matters



Faster MTTR —

Detects anomalies every 30s, diagnoses instantly — not when an engineer gets the page



AI-Grade Reasoning —

One Claude API call per incident with full network context: root cause + recommended actions



Safe Autonomy —

4-tier approval ladder: auto-executes safe changes, escalates high-blast-radius actions to humans



Adaptive Learning —

Thresholds adjust based on outcomes — catches issues sooner, reduces false positives over time

What NOC Engineers Are Up Against

User Context

- ISP backbone: Delhi ↔ Mumbai, 10 routers, 14 links
- 2.7M customers across 2 cities depend on uptime
- Small NOC team — on-call 24/7, high cognitive load
- Mix of automated alerts and manual triage workflows
- Fiber cuts, DDoS, BGP leaks can strike at any time

Key Pain Points

- ⚠ **Alert Storm Fatigue:**
Hundreds of alerts per incident — impossible to triage fast enough
- ⚠ **Slow Diagnosis:**
Engineers manually correlate telemetry, BGP logs, topology — minutes lost
- ⚠ **Inconsistent Remediation:**
Action quality varies by shift, experience, and time of night
- ⚠ **No Autonomous Safety Net:**
Low-risk actions still require human approval — bottleneck on MTTR
- ⚠ **No Learning Loop:**
Alert thresholds stay static even as the network evolves

"Every minute of undetected degradation affects thousands of customers — and engineers are playing catch-up by the time alerts fire."

How the NOC Agent Works



Core Approach — Single-Shot Claude Pattern

Pre-fetch ALL network context via HTTP → pack into ONE prompt → ONE Claude API call per incident → parse JSON actions → create decision records



Observe & Detect

Every 30s: fetch telemetry, check CPU/memory/latency/packet-loss thresholds — no API cost



Diagnose with Claude

One API call with topology, BGP state, 30-min telemetry history → root cause + recommended actions as JSON



Safe Autonomy Tiers

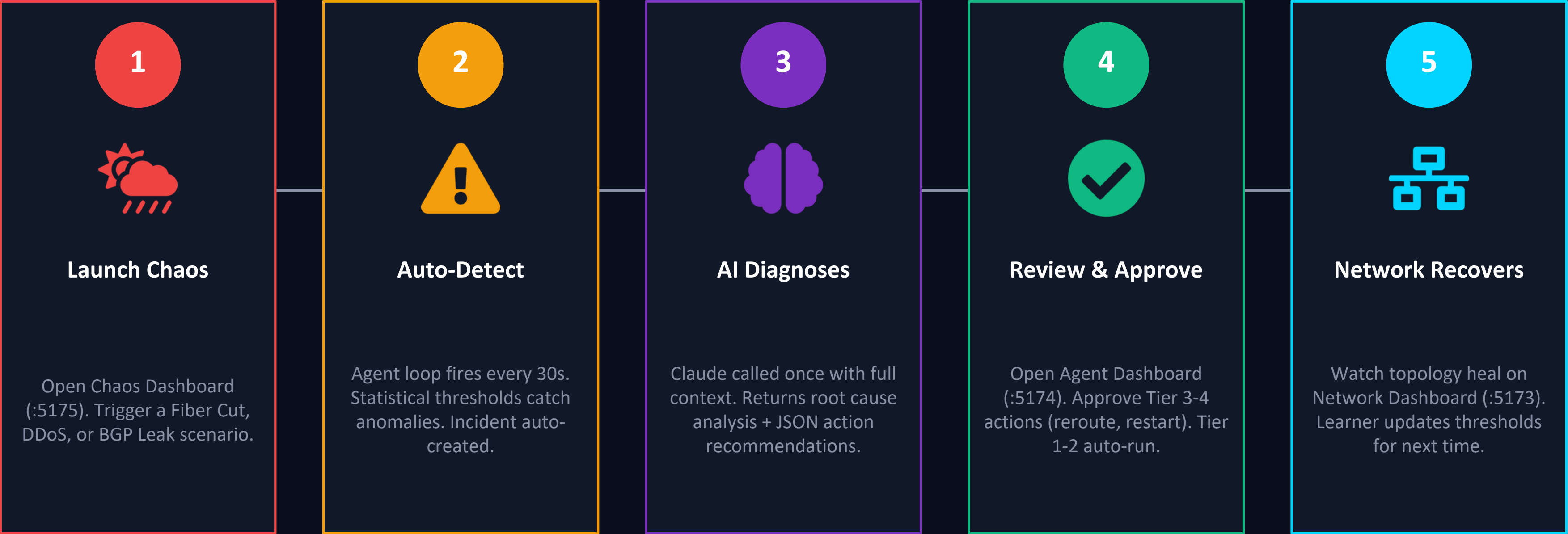
Tier 1-2 auto-execute (alerts, rate-limiting). Tier 3-4 require human approval (rerouting, device restarts)



Adaptive Learning

Successful actions lower detection thresholds (catch sooner); failures raise them (reduce noise)

From Chaos to Resolution



Outcome: Incident resolved autonomously (or with single human approval click). Network health restored. Thresholds updated.

Architecture & Experience Principles

3 React Dashboards

:5173 · :5174 · :5175

- Network Topology Map
- Agent Incidents & Approvals
- Chaos Scenarios & Impact

2 FastAPI Backends

:8000 · :8001

- Network Simulator + Chaos Eng.
- NOC Agent + Claude Reasoner
- WebSocket real-time events

Data Layer

PostgreSQL + Redis

- TimescaleDB hypertables
- Redis pub/sub event bus
- Shared schema, no conflicts

Experience Principles

Human-in-the-Loop

High-blast-radius actions always require human approval — AI recommends, humans decide

Predictable Cost

1 Claude API call per incident. Observe/detect loop is zero-AI-cost statistical checking

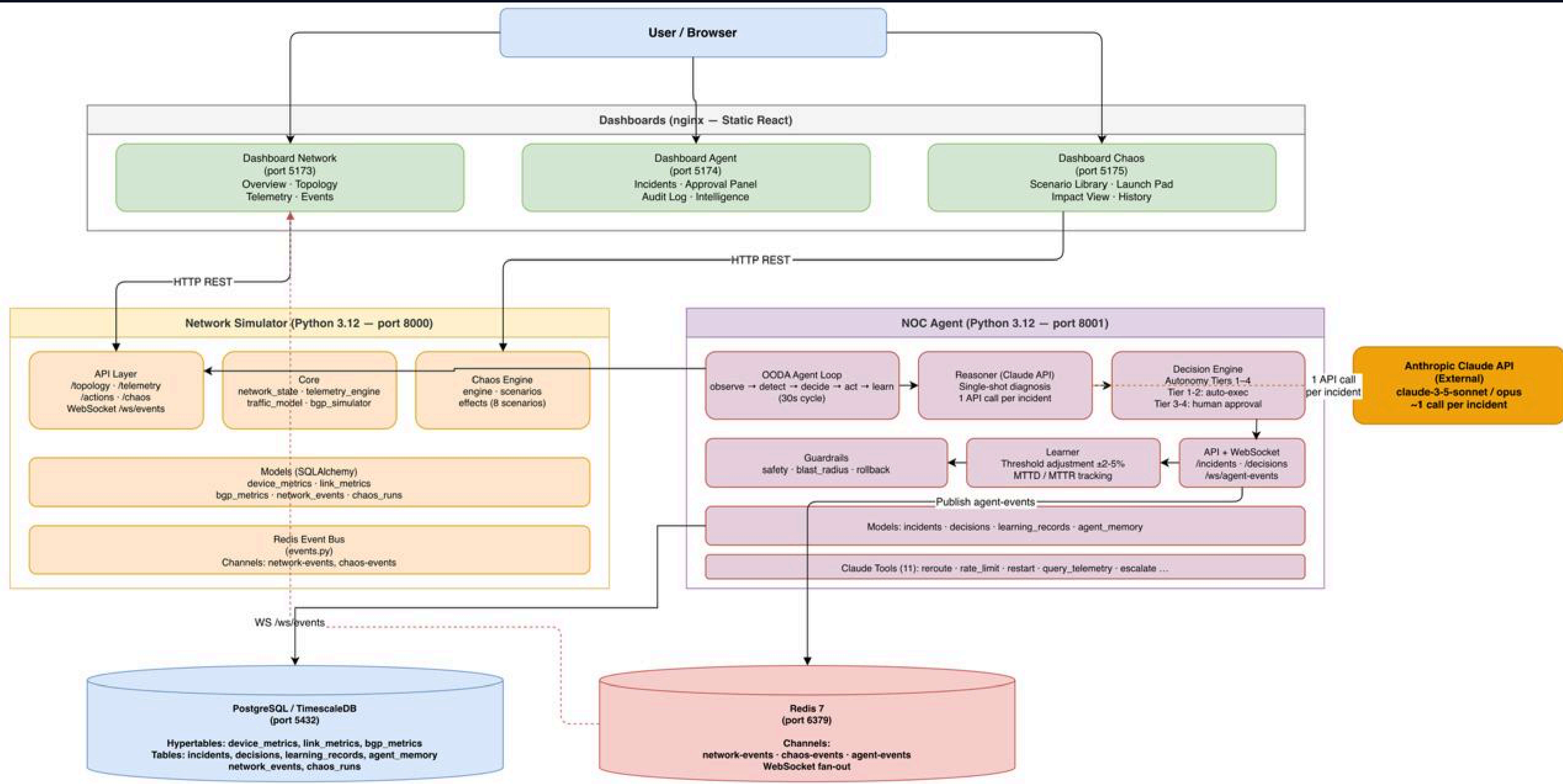
Full Observability

Every AI reasoning trace, decision, and outcome logged. Audit trail for every action taken

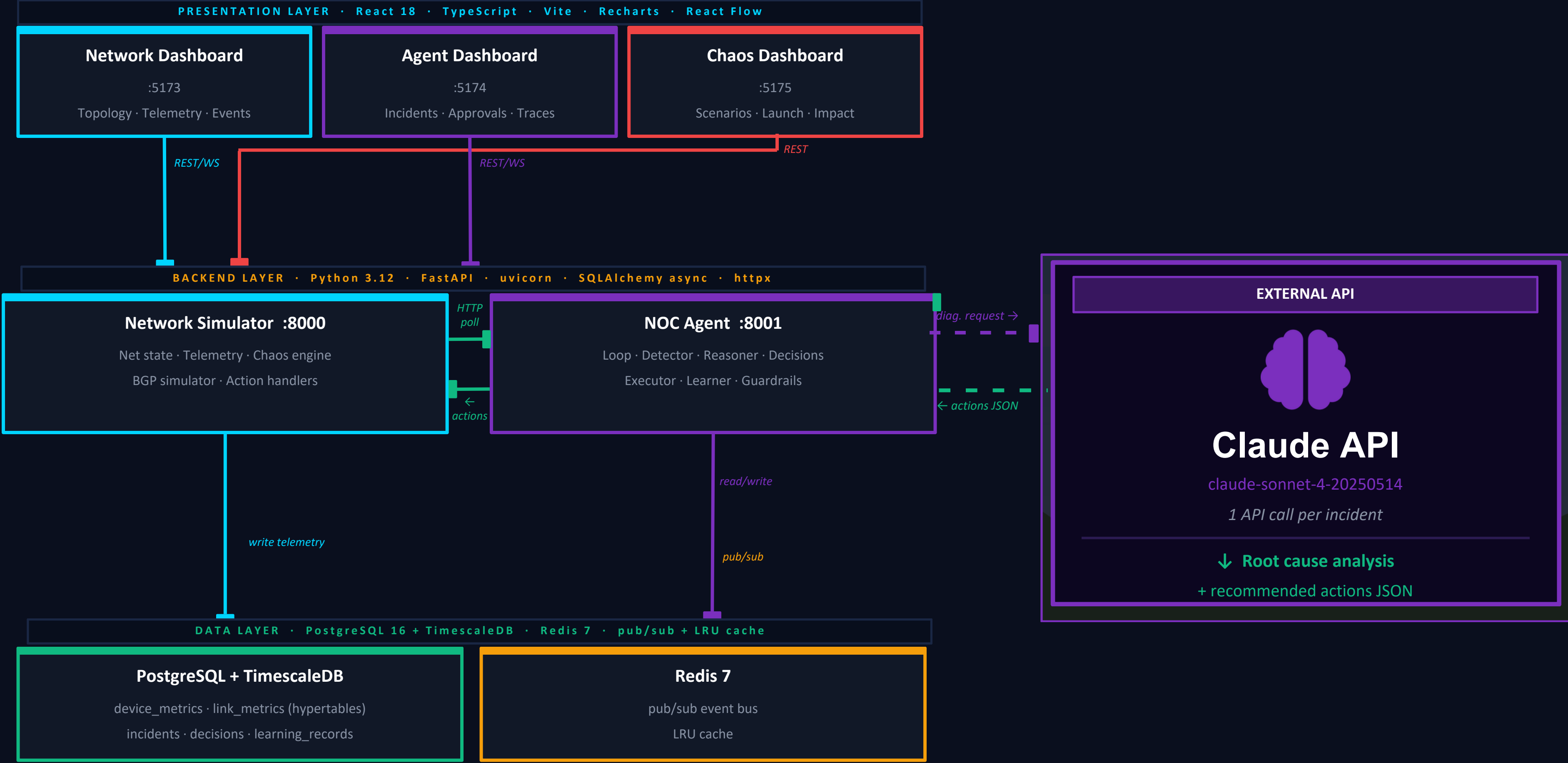
Graceful Degradation

Agent runs in observe-only or supervised mode. Guardrails rollback failed actions automatically

Block Diagram



How Data Moves Through the System



Feasibility, Scope & Vision



Feasibility

- **Fully dockerized**

7 containers, `docker compose up` — zero local dependencies beyond Docker + API key

- **Cost-controlled AI**

1 API call/incident keeps Claude costs predictable and budget-friendly

- **Safe by default**

Blast-radius guardrails, rollback mechanisms, audit trail built into every action path

- **Production patterns**

FastAPI async, TimescaleDB, Redis pub/sub — production-grade tech stack throughout

- **Observable & debuggable**

Structured JSON logs, WebSocket real-time events, full AI reasoning traces stored



Future Scope



- **Multi-turn diagnosis**

SOON

Claude follow-up questions for ambiguous incidents — deeper reasoning with more API calls when warranted



- **Predictive maintenance**

SOON

Detect slow memory leaks and link degradation before customer-impacting thresholds are hit



- **Multi-city topology**

SOON

Extend simulator to Chennai, Kolkata — model inter-city BGP peering at scale



- **Fine-tuned diagnosis**

SOON

Train a specialized model on accumulated incident history + outcomes for faster, cheaper inference



- **Real network integration**

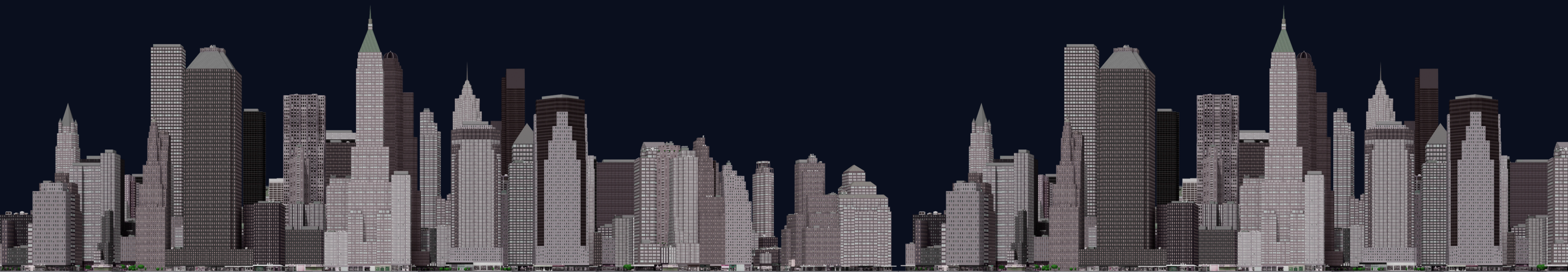
SOON

Swap simulator APIs for SNMP, gNMI, and vendor APIs to operate on live infrastructure



Autonomous NOC Agent — AI that watches your network so your engineers can focus on what matters.

THANK YOU!!



CHAOS LAB
Network Resilience

Scenario Library

Launch Pad

Impact View

History

AUTONOMOUS NETWORK // CHAOS LABORATORY

SYSTEM ONLINE

Scenario Library
8 chaos scenarios available

SEARCH

CATEGORY: All Link Device Security Routing SEVERITY: All Medium High Critical Emergency

✂️

CRITICAL

LINK

Fiber Cut

Simulates a complete fiber optic cable cut on a primary link, causing immediate traffic rerouting and potential service degradation.

EXPECTED EFFECTS

- Complete link failure
- Traffic rerouting via backup paths
- Increased latency on alternate routes
- +1 more

Primary linkConnected routersDependent services

📶

MEDIUM

LINK

Gradual Degradation

Slowly degrades a link with increasing latency and packet loss, simulating cable damage or environmental interference.

EXPECTED EFFECTS

- Steadily rising latency
- Increasing packet loss
- Quality of service degradation
- +1 more

Target linkEnd-to-end path qualitySLA metrics

⚡

EMERGENCY

SECURITY

DDoS Attack

Floods a target device with massive traffic volumes, overwhelming interfaces and causing service disruption.

EXPECTED EFFECTS

- Interface saturation
- CPU/memory spike on target
- Collateral congestion on upstream links
- +1 more

Target deviceUpstream linksConnected customers

💻

CRITICAL

DEVICE

Device Failure

Simulates a complete device failure with graceful shutdown, causing all connected links to go down.

EXPECTED EFFECTS

- All device ports go down
- Connected links fail
- Routing reconvergence
- +1 more

Target deviceAll connected linksRouting domain

🔄

EMERGENCY

ROUTING

BGP Route Leak

Injects unauthorized prefix advertisements into the BGP routing table, simulating a route leak from a peering partner.

EXPECTED EFFECTS

- Route table pollution
- Traffic misdirection
- Potential loops
- +1 more

BGP peering routerGlobal routing tableInternet-facing services

🚦

HIGH

LINK

Congestion Cascade

Triggers a cascading congestion event starting from one link and spreading to adjacent links as traffic redistributes.

EXPECTED EFFECTS

- Initial link congestion
- Cascading overload on neighbors
- Progressive quality degradation
- +1 more

Initial linkAdjacent linksRegional network segment

💾

HIGH

DEVICE

Memory Leak

Simulates a software bug causing steadily increasing memory consumption until the device crashes at threshold.

EXPECTED EFFECTS

- Gradual memory increase
- Performance degradation
- Process restarts
- +1 more

Target deviceDevice processesConnected services

📶

MEDIUM

LINK

Flapping Link

Rapidly toggles a link up and down at regular intervals, causing routing instability and excessive control plane load.

EXPECTED EFFECTS

- Rapid link state changes
- Routing table instability
- Control plane CPU spikes
- +1 more

Flapping linkConnected routersRouting protocol stability

Chaos Engineering v1.0

localhost:5175/scenarios

←→↻🏠📍localhost:5174/approvals

🔍☆📁⬇️👤Relaunch to update⋮

📡AI NOC Engineer
AUTONOMOUS OPS

🖱️Command Center

⚠️Incidents

🛡️Approval Panel

👤Field Tasks

🕒Agent Intelligence

📄Audit Log

● Agent Connected

<

Approval Panel

Autonomous Network Operations Center

del Live● NTP SyncedIST 11:25:55

🔴 Listening for events...

🚨4
ACTIVE INCIDENTS

✅ ACTIONS EXECUTED — REVIEW OUTCOME

Mar 8, 2026, 07:49 AM

INC-7CRITICAL

CRITICAL: Link Down [congestion_cascade]

↔️ AFFECTED LINKS

link-del-c1-en

ANOMALY DETECTION

Link link-del-c1-en is DOWN

🤖 AI DIAGNOSIS

ROOT CAUSE ANALYSIS

The root cause is a **congestion cascade failure** triggered by the complete failure of link-del-c1-en (core-delhi-1 to edge-delhi-north). This link shows 100% packet loss, indicating a hard failure. The cascade is evident from:

- Primary trigger:** link-del-c1-en is completely down (100% packet loss)
- Cascade effects:** 4 other links now degraded due to traffic redistribution
- Historical context:** Recent repeated failures of link-del-mum-primary have reduced network redundancy
- Traffic redistribution strain:** Remaining Delhi edge connectivity (link-del-c1-es, link-del-c2-en, link-del-c2-es) is absorbing the failed link's traffic, causing degradation

The 31.93% utilization reading for the failed link is likely stale data - the 100% packet loss confirms complete failure.

CUSTOMER IMPACT

- Immediate impact:** ~150,000 customers in North Delhi losing connectivity
- Secondary impact:** ~300,000 additional customers experiencing degraded performance due to congestion on remaining links
- Business services:** Critical enterprise customers in North Delhi business district completely offline
- Network-wide:** 7.6% average packet loss indicates widespread service degradation

RECOMMENDED ACTIONS

3538

```
{
  {
    "action_type": "execute_reroute",
    "description": "Emergency reroute North Delhi traffic through backup paths",
    "parameters": {
      "from_link": "link-del-c1-en",
      "to_link": "link-del-c2-en",
      "reason": "Complete failure of primary North Delhi edge link causing customer outage"
    },
    "reasoning": "Immediately restore connectivity for North Delhi customers by redistributing traffic to functioning links",
    "expected_result": "Restore service for ~150K customers with temporary 15-20ms latency increase",
    "confidence": 0.9,
    "blast_radius": 150000
  },
  {
    "action_type": "apply_rate_limit",
    "description": "Apply rate limiting on overloaded Delhi links to prevent further cascade",
    "parameters": {
      "target": "link-del-c1-es",
      "limit_mbps": 32000,
      "duration_minutes": 120,
      "reason": "Prevent congestion cascade on remaining Delhi edge links"
    },
    "reasoning": "Stabilize network by controlling traffic flow on degraded links until permanent fix",
    "expected_result": "Reduce packet loss from 7.6% to ~2%, stabilize remaining links",
    "confidence": 0.85,
    "blast_radius": 300000
  },
  {
    "action_type": "dispatch_field_tech",
    "description": "Emergency fiber splice repair for link-del-c1-en",
```

Network Topology

- Overview

- Devices

- ## 🔗 Interfaces

- Routing

- Ln Traffic

- Alerts

- Metrics

-  Compliance

- Subscribers

-  Audit Trail

- System Status
 - Simulator Connected
 - MT P Synced

localhost:5173/routing

